



1

Partitioning (Centeroid-Based) Algorithms

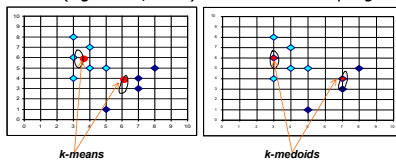
- Construct a partition of a database D of n objects into a set of k clusters
- Given a k , find a partition of k clusters that optimizes the chosen partitioning criterion
 - k -means (MacQueen'67)
 - Each cluster is represented by the center of the cluster
 - A Euclidean Distance based method, mostly used for interval/ratio scaled data
 - k -medoids
 - Each cluster is represented by one of the objects in the cluster
 - For categorical data

2

2

k -Medoids Clustering Method

- k -medoids: Find k representative objects, called *medoids*
- PAM (Partitioning Around Medoids, 1987)
- CLARA (Kaufmann & Rousseeuw, 1990)
- CLARANS (Ng & Han, 1994): Randomized sampling



3

PAM (Partitioning Around Medoids) (1987)

- PAM (Kaufman and Rousseeuw, 1987)
- Arbitrarily choose k objects as the initial medoids
- Until no change, do
 - (Re)assign each object to the cluster with the nearest medoid
 - Improve the quality of the k -medoids
 - (Randomly select a nonmedoid object, O_{random})
 - compute the total cost of swapping a medoid with O_{random}
- Work for small data sets (100 objects in 5 clusters)
- Not efficient for medium and large data sets

4

4

Swapping Cost

- For each pair of a medoid o and a non-medoid object h , measure whether h is better than o as a medoid
- Use the squared-error criterion

$$E = \sum_{i=1}^k \sum_{p \in C_i} d(p, o_i)^2$$

- Compute $E_h - E_o$
- Negative: swapping brings benefit
- Choose the minimum swapping cost

5

5

Four Swapping Cases

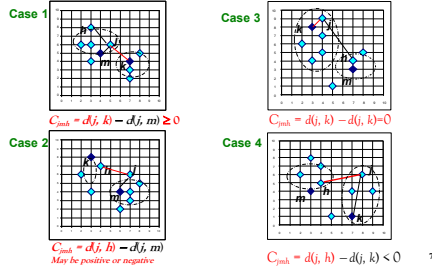
- When a medoid m is to be swapped with a non-medoid object h , check each of other non-medoid objects j
 - j is in cluster of $m \Rightarrow$ reassign j
 - Case 1: j is closer to some k than to h ; after swapping m and h , j relocates to cluster represented by k
 - Case 2: j is closer to h than to k ; after swapping m and h , j is in cluster represented by h
 - j is in cluster of some k , not $m \Rightarrow$ compare k with h
 - Case 3: j is closer to some k than to h ; after swapping m and h , j remains in cluster represented by k
 - Case 4: j is closer to h than to k ; after swapping m and h , j is in cluster represented by h

6

6

PAM Clustering: Total swapping cost

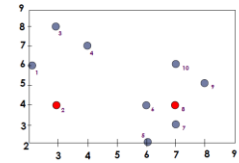
$$TC_{mh} = \sum_j C_{jmh}$$



K-Medoids Method: Example

Data Objects

	A ₁	A ₂
O ₁	2	6
O ₂	3	4
O ₃	3	8
O ₄	4	7
O ₅	6	2
O ₆	6	4
O ₇	7	3
O ₈	7	4
O ₉	8	5
O ₁₀	7	6



Goal: create two clusters

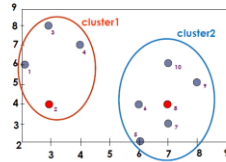
Choose randomly two medoids

O₂ = (3, 4)
O₈ = (7, 4)

K-Medoids Method: Example

Data Objects

	A ₁	A ₂
O ₁	2	6
O ₂	3	4
O ₃	3	8
O ₄	4	7
O ₅	6	2
O ₆	6	4
O ₇	7	3
O ₈	7	4
O ₉	8	5
O ₁₀	7	6



→Assign each object to the closest representative object

→Using L1 Metric (Manhattan), we form the following clusters

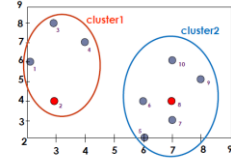
Cluster1 = {O₁, O₂, O₃, O₄}

Cluster2 = {O₅, O₆, O₇, O₈, O₉, O₁₀}

K-Medoids Method: Example

Data Objects

	A ₁	A ₂
O ₁	2	6
O ₂	3	4
O ₃	3	8
O ₄	4	7
O ₅	6	2
O ₆	6	4
O ₇	7	3
O ₈	7	4
O ₉	8	5
O ₁₀	7	6



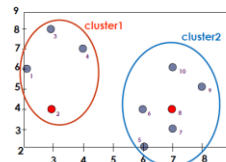
→Compute the absolute error criterion [for the set of Medoids {O₂, O₈}]

$$E = \sum_{i=1}^k \sum_{j \in C_i} p - q_i | = |q_1 - o_2| + |q_3 - o_2| + |q_4 - o_2| + |q_5 - o_8| + |q_6 - o_8| + |q_7 - o_8| + |q_8 - o_8| + |q_9 - o_8| + |q_{10} - o_8|$$

K-Medoids Method: Example

Data Objects

	A ₁	A ₂
O ₁	2	6
O ₂	3	4
O ₃	3	8
O ₄	4	7
O ₅	6	2
O ₆	6	4
O ₇	7	3
O ₈	7	4
O ₉	8	5
O ₁₀	7	6



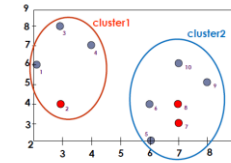
→The absolute error criterion [for the set of Medoids {O₂, O₈}]

$$E = (3 + 4 + 4) + (3 + 1 + 1 + 2 + 2) = 20$$

K-Medoids Method: Example

Data Objects

	A ₁	A ₂
O ₁	2	6
O ₂	3	4
O ₃	3	8
O ₄	4	7
O ₅	6	2
O ₆	6	4
O ₇	7	3
O ₈	7	4
O ₉	8	5
O ₁₀	7	6



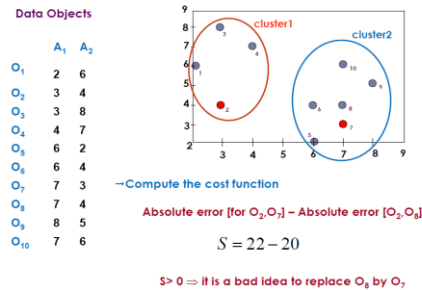
→Choose a random object O₇

→Swap O₈ and O₇

→Compute the absolute error criterion [for the set of Medoids {O₂, O₇}]

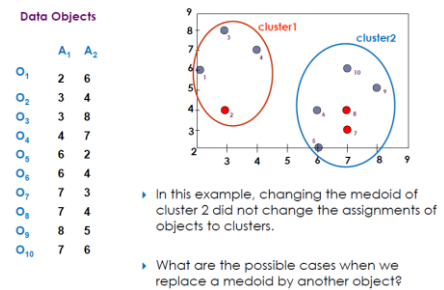
$$E = (3 + 4 + 4) + (2 + 2 + 1 + 3 + 3) = 22$$

K-Medoids Method: Example



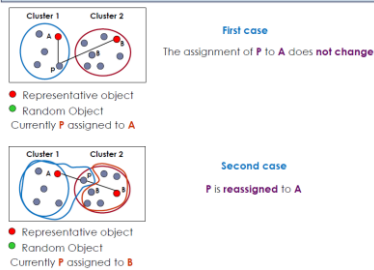
13

K-Medoids Method



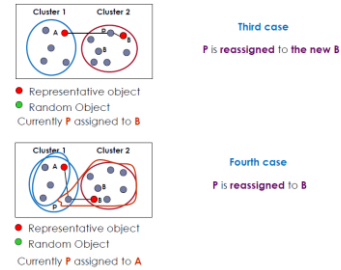
14

K-Medoids Method



15

K-Medoids Method



16

K-Medoids Algorithm(PAM)

PAM : Partitioning Around Medoids

- ▶ **Input**
 - K: the number of clusters
 - D: a data set containing n objects
- ▶ **Output:** A set of k clusters
- ▶ **Method:**
 - (1) Arbitrary choose k objects from D as representative objects (seeds)
 - (2) **Repeat**
 - (3) Assign each remaining object to the cluster with the nearest representative object
 - (4) For each representative object O_j
 - (5) Randomly select a non representative object O_{random}
 - (6) Compute the total cost S of swapping representative object O_j with O_{random}
 - (7) if $S < 0$ then replace O_j with O_{random}
 - (8) **Until** no change

17

K-Medoids Properties(k-medoids vs.K-means)

- ▶ The complexity of each iteration is $O(k(n-k)^2)$
- ▶ For large values of n and k, such computation becomes very costly
- ▶ **Advantages**
 - K-Medoids method is more robust than k-Means in the presence of noise and outliers
- ▶ **Disadvantages**
 - K-Medoids is more costly than the k-Means method
 - Like k-means, k-medoids requires the user to specify k
 - It does not scale well for large data sets

18

Strength and Weakness of PAM

- PAM is more robust than k-means in the presence of outliers because a medoid is less influenced by outliers or other extreme values than a mean
- PAM works efficiently for small data sets but does not scale well for large data sets
- $O(k(n-k)^2)$ for each iteration
where n is # of data objects, k is # of clusters

19